STSMN - ESP32 wiring guide

Smart Transformer Safety & Monitoring Network. Ye guide beginner ke liye hai: pehle saaman, phir diagram, phir pin-se-pin wiring.

Kul 8 modules: 6 sensor modules (1 Temperature, 2 Current, 3 Voltage, 4 Smoke, 5 Flame, 6 Water level) + 7 GPS + 8 SIM (SMS). Saath mein 3 output: Relay, Buzzer, 3 LED.

Saaman ki list (asli components)

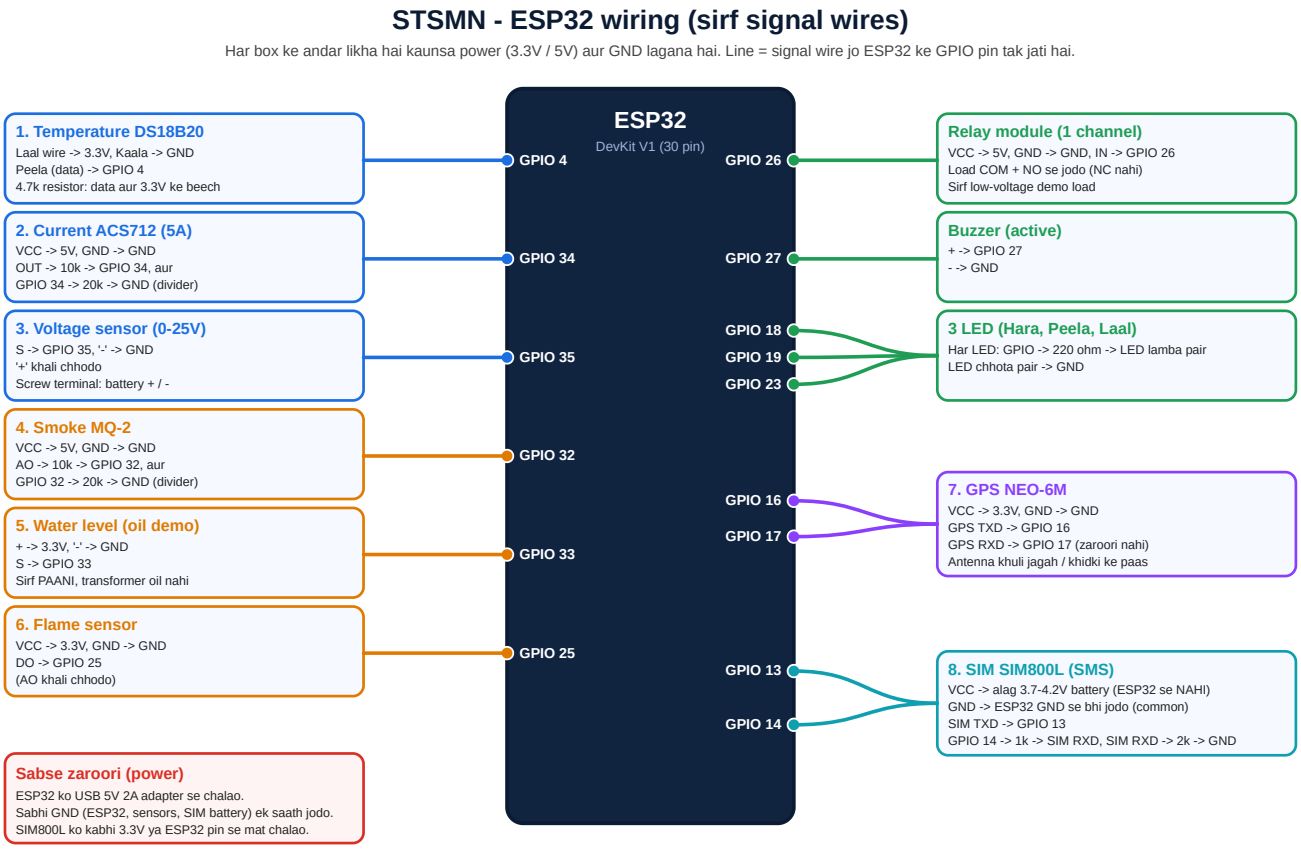
#	Component (market mein naam)	Kaam	Kitne
1	ESP32 DevKit V1 (30 pin, ESP32-WROOM-32)	Dimaag: sab sensors padhta hai, relay chalata hai, Wi-Fi deta hai	1
2	DS18B20 waterproof temperature probe (3 wire)	Temperature	1
3	ACS712 5A current sensor module	Current (Ampere)	1
4	Voltage sensor module 0-25V (FZ0430 type)	Voltage	1
5	MQ-2 smoke / gas sensor module	Dhuaan	1
6	Flame sensor module (IR, 3 ya 4 pin)	Aag	1
7	Water level sensor (analog, 3 pin)	Oil level ka demo (sirf paani)	1
8	NEO-6M GPS module + antenna	Asli location (latitude/longitude)	1
9	SIM800L GSM module + antenna + SIM card	SMS alert bhejna	1
10	1 channel relay module 5V	Supply kaatna (cutoff)	1
11	Active buzzer	Alarm awaaz	1
12	LED hara, peela, laal	Status light	3
13	Resistor 220 ohm	Har LED ke saath	3
14	Resistor 10k ohm / 20k ohm	ACS712 aur MQ-2 ke divider (har ek ke liye 10k + 20k)	2 + 2
15	Resistor 1k ohm / 2k ohm	SIM800L RX divider	1 + 1
16	Resistor 4.7k ohm	DS18B20 pull-up	1
17	Capacitor 1000 uF 10V (ya zyada)	SIM800L ke paas power ko steady rakhta hai	1
18	18650 Li-ion cell + holder (ya 4V 2A buck converter)	SIM800L ka alag power	1
19	Breadboard + male-female / male-male jumper wires	Jodne ke liye	-
20	USB 5V 2A adapter + micro-USB cable	ESP32 ka power	1

Pehle safety (zaroori)

- Ye **sirf low-voltage demo** hai (battery / 3.7V - 12V). **230V mains ya asli transformer se kabhi mat jodo.**
- Wiring karte waqt ESP32 ka USB nikalo. Pehle sab jodo, phir jaancho, phir power do.
- Relay ke load wale taar COM aur **NO** mein lagao (NC mein nahi): ESP32 band ho to supply kat jaaye.

- SIM800L ka power **ESP32 ke 3.3V ya 5V pin se mat lo**: usse bahut current chahiye, ESP32 reset hoga.
- Flame test ke time bade ki nigrani mein raho. Lighter sensor se 20 cm door rakho.

Diagram



Step by step wiring (pin se pin)

Rang ka matlab: **laal = power**, **kaala = GND**, baaki = signal. Har module ko isi order mein jodo.

Module	Module ka pin	Kahan lagana hai
1. DS18B20	Laal (VCC)	ESP32 **3V3**
	Kaala (GND)	ESP32 **GND**
	Peela (DATA)	ESP32 **GPIO 4** + 4.7k resistor DATA aur 3V3 ke beech
2. ACS712	VCC	ESP32 **VIN (5V)**
	GND	ESP32 **GND**
	OUT	10k resistor -> **GPIO 34**. GPIO 34 se 20k resistor -> GND
	IP+ / IP-	Load ke series mein (battery + se load tak, beech mein sensor)
3. Voltage sensor	S	ESP32 **GPIO 35**
	-	ESP32 **GND**
	+ (sensor side)	Khali chhodo
	Screw terminal VCC / GND	Naap-ne wali battery + aur - (15V se zyada nahi)

4. MQ-2	VCC	ESP32 **VIN (5V)**
	GND	ESP32 **GND**
	AO	10k resistor -> **GPIO 32**. GPIO 32 se 20k resistor -> GND
	DO	Khali chhodo
5. Water level	+ (VCC)	ESP32 **3V3**
	- (GND)	ESP32 **GND**
	S	ESP32 **GPIO 33**
6. Flame	VCC	ESP32 **3V3**
	GND	ESP32 **GND**
	DO	ESP32 **GPIO 25**
7. GPS NEO-6M	VCC	ESP32 **3V3**
	GND	ESP32 **GND**
	TXD	ESP32 **GPIO 16**
	RXD	ESP32 **GPIO 17** (na jodo to bhi chalega)
8. SIM800L	VCC	**Alag battery +** (3.7-4.2V). 1000 uF capacitor VCC-GND ke beech
	GND	Battery - **aur** ESP32 GND (dono GND ek hon)
	TXD	ESP32 **GPIO 13**
	RXD	ESP32 **GPIO 14** se 1k resistor -> SIM800L RXD. SIM800L RXD se 2k resistor -> GND
	Antenna + SIM card	Antenna lagao, SIM (SMS wala, PIN lock band) lagao
Relay	VCC / GND / IN	ESP32 **VIN** / **GND** / **GPIO 26**
	COM + NO	Load ka taar (low voltage)
Buzzer	+ / -	**GPIO 27** / GND
LED hara	lamba pair -> 220 ohm	**GPIO 18**; chhota pair -> GND
LED peela	lamba pair -> 220 ohm	**GPIO 19**; chhota pair -> GND
LED laal	lamba pair -> 220 ohm	**GPIO 23**; chhota pair -> GND

Code ESP32 mein kaise daalein

1. Arduino IDE 2.x kholo. *Boards Manager* se **ESP32 by Espressif Systems** install karo.
2. *Library Manager* se **OneWire** aur **DallasTemperature** install karo.
3. File `stsmn_single.ino` poori kholo. Upar `WIFI_AP_PASSWORD` apna naya password (8-63 akshar) karo.
4. `ALERT_PHONE_NUMBER` mein apna number daalo, country code ke saath, jaise `+919876543210`. Ye na badlo to SMS band rahega.
5. Board: **ESP32 Dev Module**. Sahi COM port chuno. Upload dabao.
6. Serial Monitor (115200) kholo. Dikhna chahiye: `SMS alerts: ON` aur kuch second baad `SIM module ready`.

Jaanch (test) karne ka tareeka

Kya jaanchna hai	Kaise	Sahi result
Wi-Fi + app	Phone mein Wi-Fi **STSMN-TR001**, app mein 192.168.4.1	App mein live data
GPS	Module khuli jagah / khidki par rakho, 1-10 minute ruko. Browser: 192.168.4.1/api/health	`gpsFix:true`, lat/lng aapke shehar ke aas-paas
SIM signal	Wahi /api/health	`gsmReady:true`, `gsmRegistered:true`, `gsmSignal` 10 se upar
SMS alert	Bade ki nigrani mein lighter ki lau flame sensor ke paas (20 cm) lao	Relay kat jaata hai, buzzer bajta hai, kuch second mein SMS aata hai
Smoke	Agarbatti ka dhuaan MQ-2 ke paas (pehle 60 second warm-up)	Smoke true

Aam dikkatein

Dikkat	Kaaran / hal
SIM module not answering	Power kam hai. Alag battery + 1000 uF capacitor lagao. GND common karo. TX/RX ulta to nahi (module TXD -> ESP GPIO 13).
SMS nahi aata, gsmRegistered false	SIM800L sirf 2G chalata hai. Aapke area mein 2G network (Airtel / Vi / BSNL) hona chahiye; Jio SIM nahi chalegi.
gpsFix false	Antenna ko khuli jagah rakho. Pehli baar fix mein 10 minute tak lag sakte hain. TXD -> GPIO 16 check karo.
ESP32 baar baar restart	USB adapter kamzor hai ya SIM800L ESP32 ke power se juda hai.
Current / voltage galat	Multimeter se milaakar `VOLT_CAL_FACTOR` aur ACS712 ka divider check karo (config.h).

Sach batana (limits)

- Ye code aur wiring **asli hardware par abhi test nahi hue**. Sirf computer par logic test aur syntax check hue hain. Pehli baar chalate waqt ek-ek module alag jaancho.
- SMS mein location sirf tab aati hai jab GPS ka asli fix ho; warna likha aata hai *GPS: no fix*. Koi nakli location nahi bheji jaati.
- Sensor ki limits (config.h) demo values hain; aapke sensors ke hisaab se calibrate karni hongi.
- Ye educational demo hai, certified protection device nahi.